Bandana Dhakal

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Professional Summary

BSC Physics graduate from Shaheed Smriti Multiple Campus, Institute of Science and Technology (IOST), Tribhuvan University, Nepal, with a strong academic record and research experience in household energy consumption, fuel stacking, and sustainable energy transitions. Ranked first in the Physics programme throughout all four academic years and received a departmental merit scholarship. Experienced in quantitative data analysis, statistical modelling, and scientific computing using Python and MATLAB.

Experience

Lead Author and Thesis Researcher
March 2026 to August 2026
Shaheed Smriti Multiple Campus, IOST, Tribhuvan University - Chitwan

- Led undergraduate research investigating household energy consumption, fuel stacking, and electricity adoption across 201 households in two semi-urban settlements in Chitwan, Nepal.
- Conducted statistical analysis using Python (pandas, NumPy, SciPy, Matplotlib) to derive insights from data, including descriptive statistics, logistic regression, and sensitivity analysis.
- Developed computational models to estimate useful cooking energy consumption, determining average consumption of 246 MJ per household per month from LPG.
- Applied scientific computing and quantitative research methods to interpret findings and effectively communicate research outcomes.

Education

Bachelor of Science (BSc) in Physics -Completed October 2026
Shaheed Smriti Multiple Campus, Institute of Science and Technology (IOST), Tribhuvan University - Chitwan, Nepal

- Ranked first in the Physics department throughout all four years of the Bachelor of Science programme (2022–2026).
- Completed a final-year research thesis titled "Fuel Stacking and Electricity Use in Two Semi-Urban Settlements in Nepal," achieving a score of 92, the highest among the graduating cohort.
- Conducted quantitative research involving 201 households, applying Python programming, statistical analysis, logistic regression, and numerical modelling.
- Developed expertise in classical mechanics, quantum physics, electromagnetism, thermodynamics, statistical physics, and mathematical physics.
- Awarded a full departmental Academic Excellence Scholarship throughout the four-year undergraduate programme in recognition of exceptional academic performance.

Higher Secondary School : **+2 Science**, 2021
Nepal Higher Secondary School - Chitwan, nepal

- Graduated with a GPA of 3.43/4.00 in the Higher Secondary Education (+2 Science) programme with majors in maths and Physics
- Achieved first rank among students in the +2 Science programme, demonstrating outstanding academic performance.
- Awarded a full academic scholarship in recognition of exceptional academic achievement and first-place ranking.
- Developed a strong academic foundation in Physics, Mathematics, Chemistry, and scientific problem-solving.
- [Award] Awarded a full academic scholarship in recognition of exceptional academic achievement and first-place ranking.

Secondary Education Examination (SEE) : June 2019
Ekata Shishu Niketan - Chitwan, nepal

- Completed the Secondary Education Examination (SEE) with a GPA of 3.50 out of 4.00.

- Developed a strong academic foundation in Mathematics, Science, English, and other core subjects.
- Demonstrated academic dedication, analytical thinking, and problem-solving abilities throughout secondary education.

Skills

- Python
- LaTeX
- PowerPoint
- Scientific computing
- Mathematical modeling
- JAVA
- C
- GitHub
- Microsoft Excel and MS Word
- Computational physics
- Numerical analysis
- Data science
- HTML
- C++

References

Deepak Regmi - Shaheed Smriti Multiple Campus
B.Sc. Coordinator

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Basant Acharya
Lecturer and Thesis Supervisor

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Languages

- **Nepali** (Native)
- **English**